

Fundamental Statistical Concepts

1 Hypothesis Testing

Hypothesis testing is a formal statistical procedure used to make decisions about a population parameter based on sample data. The process begins with two competing statements: the *null hypothesis* (H_0) and the *alternative hypothesis* (H_1). The null hypothesis typically represents the status quo or no effect, while the alternative hypothesis represents a deviation or effect of interest.

A decision rule is defined in advance, specifying whether to *reject* or *not reject* the null hypothesis. This decision is based on a test statistic and its corresponding *p-value*. The p-value represents the probability of observing data at least as extreme as the sample, assuming the null hypothesis is true.

Statistical significance is determined by comparing the p-value to a predefined threshold (commonly denoted α , such as 0.05). If the p-value is smaller than α , the result is considered statistically significant, and the null hypothesis is rejected. Importantly, this does not prove the alternative hypothesis, but rather indicates that the observed data would be unlikely under the null hypothesis.

2 Confidence Intervals

A confidence interval is an interval estimate used to infer the value of a population parameter based on sample data. Instead of providing a single estimate, it provides a range of plausible values.

The confidence level (e.g., 95%) refers to the long-run frequency with which such intervals would contain the true parameter if the sampling process were repeated many times. It is crucial to note that the confidence level does *not* represent the probability that the true parameter lies within a specific computed interval.

Confidence intervals reflect uncertainty arising from sampling variability. Wider intervals indicate greater uncertainty, while narrower intervals suggest more precise estimates. The width of the interval depends on factors such as sample size and variability in the data.

3 Mean and Median

The *mean* is the arithmetic average of a dataset, calculated as the sum of all observations divided by the number of observations. The *median* is the middle value when the data are ordered from smallest to largest.

A key difference between the two measures is their sensitivity to outliers. The mean is strongly influenced by extreme values, whereas the median is more robust and less affected by such outliers.

In symmetric distributions, the mean and median are typically equal. However, in skewed distributions, they differ: in right-skewed distributions, the mean is greater than the median, while in left-skewed distributions, the mean is smaller than the median. This distinction makes the median particularly useful for describing central tendency in skewed data.

4 Variance

Variance is a measure of how spread out data points are around the mean. It is calculated as the average of the squared deviations from the mean:

$$\text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

A high variance indicates that data points are widely dispersed, while a low variance suggests that they are clustered closely around the mean.

The *standard deviation* is closely related to variance and is defined as its square root. Both measures capture the same concept of variability, but the standard deviation is often preferred because it is expressed in the same units as the original data.